

Real-Time Cyber-Physical Event Discrimination in Inverter-Dominated Microgrids: Residual-Prior Fusion, Not Topology-Awareness, Drives the Gains

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Abstract—Distinguishing a cyber false-data-injection attack (FDIA) from a legitimate physical disturbance, localizing it, and doing so within a protection-relevant time budget are usually studied separately. We study all three on a synthetic inverter-dominated microgrid with a weighted-least-squares (WLS) AC state estimator, built specifically to test whether explicit topology-relational features (neighbor- and incident-edge-comparison statistics) help. A first pass appeared to show they did: `topology_fusion` beat a prior-based baseline at both detection (balanced accuracy 0.950 vs. 0.873) and localization (top-1 accuracy 0.980 vs. 0.780), confirmed across four independent seeds. That comparison was not a fair one. Topology-relational columns are named after the residual/innovation quantities they compare (e.g. `neighbor_mean_residual`), and a name-substring-based feature-set split let roughly half of them leak, mislabeled, into what were meant to be topology-free baselines – so `topology_fusion` was effectively being compared against baselines that already contained most of its own information. Correcting this and building a genuinely topology-free ablation – residual and prior-innovation features combined, with no relational comparisons at all – matches `topology_fusion` on detection (0.945 vs. 0.943 mean balanced accuracy across 4 seeds, gap -0.003 ± 0.010) and beats it on localization (0.988 vs. 0.985, gap -0.003 ± 0.007), under two independent attack-magnitude/noise conditions; unlike the original (uncorrected) comparison, the sign of this gap does not hold a consistent direction across seeds. We report this null result as the headline finding: combining residual and prior-innovation information explains the entire measured advantage over either alone; explicit topological relational structure adds nothing further on this testbed – which also now explains why the same `topology_fusion` advantage separately failed to reproduce on a larger, meshed IEEE 14-bus network built with an independent, size-agnostic feature design. We separately profile the full decision pipeline against a one-cycle (20 ms at 50 Hz) protection-relevant computational target and find the original 43 ms pipeline slow because of an avoidable software inefficiency, not the model or the state estimator’s iteration count; fixing it brings the pipeline to 16.4/17.0/17.1 ms at the median/p95/p99. A third, independent result further scopes any deployment claim: the detector does not generalize to a completely unseen attack type (0–4% recall when one of two attack families is withheld). We think catching and correcting our own feature-labeling bug, rather than reporting the more publication-friendly original result, is the more useful contribution here.

Index Terms—false data injection attack, state estimation, cyber-physical systems, attack localization, real-time systems,

feature ablation, inverter-based resources

I. INTRODUCTION

Modernizing power grids toward higher penetrations of inverter-based resources (PV, battery storage, grid-forming converters) changes the character of protection and monitoring in two ways relevant here. First, inverters provide far less fault current than synchronous generators, which strains protection schemes designed around synchronous-machine fault signatures [1]. Second, the same measurement infrastructure used for state awareness is now also an attack surface: an adversary who can spoof a subset of sensor readings can attempt a false-data-injection attack (FDIA) against the state estimator that is, by construction, statistically indistinguishable from a legitimate measurement under a residual-based bad-data test [2].

Renewable-heavy grid infrastructure specifically is not a hypothetical target: in December 2025, a destructive cyber-attack against Poland’s energy sector – corrupted firmware forced into endless restart cycles across roughly 30 wind and solar farms and a heat-and-power plant, during a cold-weather period, attributed to infrastructure overlapping with the Dragonfly/Static Tundra threat actor – was contained before it caused a blackout affecting an estimated 500,000 people [3]. That incident was an OT/firmware-integrity attack rather than a state-estimation FDIA specifically, but it establishes the same operational stakes this paper’s threat model targets: an operator or automated scheme facing a live control-plane anomaly, under real time pressure, in exactly this class of infrastructure.

A control-room operator, or an automated protection scheme, therefore faces a three-part problem for any anomalous reading: *is this a cyberattack or a physical event, where is it*, and – if it is to inform a protective action rather than only a post-hoc forensic one – *can a decision be reached inside a protection-relevant time budget*. These three questions are usually studied separately. A large FDIA-detection literature (Section II) asks the first question, typically evaluated offline against a fixed dataset. A smaller attack-localization literature asks the second. Real-time feasibility is addressed by a much

smaller, very recent body of work – notably a 2026 benchmark [4] that found deep classifiers for this exact joint task are individually fast (sub-15 ms) but the measured end-to-end pipeline is not (50–90 ms), calling this “a critical gap between algorithmic capability and protection-grade deployment” and recommending hardware acceleration. A separate, popular line of work asks whether giving a detector explicit *topology* – not just raw sensor residuals, but how each meter’s reading compares to its electrical neighbors – improves on any of this; that question is this paper’s own starting point, and its central subject.

This paper studies all three questions on one testbed, and – in the process of doing so honestly – surfaces three corrections to its own early results that we think are worth reporting as part of the method, not edited out of the narrative:

- 1) **A near-perfect early result was wrong, and wrong for a boring reason.** An initial evaluation of topology-aware detection reported 1.000 balanced accuracy across every metric and three different model families, on a 72-sample test set. We show this was not a data-leakage bug (we checked, fixed a real but inconsequential defensive gap, and the result did not change) but simply that the attack scenarios were too easy. Hardening the attack magnitude toward the sensor noise floor initially *reversed* the finding at small sample size ($n=72$) before a larger run ($n=300$) showed the original small-sample reversal was itself noise.
- 2) **The resulting “topology helps” finding did not survive its own ablation.** `topology_fusion` was compared against `residual_only` and `prior_only`, but not against a set that combines the two *without* any relational features – the actual, fair control. Building that control exposed a feature-labeling bug: explicit topology-relational columns are named after the residual/innovation quantities they compare (e.g. `neighbor_mean_residual`), so a name-substring-based split had silently let about half of them leak into what were meant to be topology-free baselines. Once corrected, the genuinely topology-free `residual_plus_prior` ablation matches or beats `topology_fusion`, across two stress conditions and four independent seeds (Section IV).
- 3) **The pipeline-latency gap reported in concurrent work is, at least here, mostly fixable software engineering, not an algorithmic ceiling.** We independently reproduce the qualitative finding of [4] (fast model, slow pipeline) and then localize the actual bottleneck: not the classifier, not the state estimator’s Newton-type iteration count, but a measurement-table object being rebuilt from scratch through a general-purpose element-creation API every cycle, when its structure never changes. Fixing that one thing accounts for most of the gap closed.

Contributions:

- A methodologically rigorous negative result: once a feature-labeling bug is corrected and a fair, same-

information ablation is built, explicit topology-relational features measure no benefit over combining residual and prior-innovation features for cyber-physical event detection or localization – confirmed across two stress conditions and four independent seeds, and independently cross-validated at a second network scale with an unrelated feature-engineering design (Section IV, Section VII), together with a structural, quantitative explanation for why (Section IV).

- A root-cause profiling methodology for the “fast model, slow pipeline” gap identified in prior work, and a demonstrated fix that closes most of it without hardware acceleration (Section V).
- A further explicit negative result reported as a scope condition rather than omitted: no zero-day attack-type generalization (Section VI).

II. RELATED WORK

Foundational FDIA and stealth constructions. The FDIA threat model used throughout this paper – an attacker with knowledge of the network model constructing a measurement perturbation that leaves the WLS residual unchanged – follows Liu, Ning & Reiter’s foundational construction [2] and its AC-consistent extensions [5]. A growing subline constructs such attacks with *less* attacker knowledge than the original construction assumes: without line parameters [6], or fully “blind” to topology using generative or physics-informed models [7], [8] – all harder threat models than the one used here and natural next stress conditions for this project’s own attack generator. In the opposite direction, low-magnitude attacks close to the sensor noise floor – the specific regime Section IV’s hardened attack magnitude targets – have their own dedicated detection literature [9], worth comparing against directly in future work.

Deep-learning FDIA detection. A substantial literature applies CNN/LSTM/transfer-learning architectures to FDIA detection [10]–[12], generally evaluated as a binary classification task on a fixed attack distribution. This project’s `residual_only` and `prior_only` baselines sit in this tradition; its contribution is isolating what *explicit topology-relational* features specifically add beyond a fair combination of the two, under a stress-tested (not default-easy) attack distribution.

Topology-aware / graph-based state estimation and detection. The closest published framing to this paper’s central question is topology-aware graph neural network (GNN) based state estimation [13]–[15], which use graph convolution/attention to exploit network structure, generally reporting improvements over non-topology baselines without, in the abstracts available to us, an ablation isolating topological structure from simply having more information combined. This paper’s corrected Section IV finding – that explicit topology-relational features measure no benefit over a simpler residual-plus-prior combination, on either of two tested network scales – is offered as a concrete methodological caution for that line of work: a positive result for a “topology-aware” architecture should be checked against a same-information,

non-topological ablation before the gain is attributed to topology specifically, since feature/architecture co-variation can otherwise produce exactly the kind of artifact this paper found, and corrected, in itself.

Attack/event localization. Localization work spans capsule networks for load-altering (not FDIA) attacks evaluated across 14/39/57-bus systems [16], moving-target-defense-assisted localization [17], and online detection-plus-localization schemes [18]. None of the abstracts available to us report a localization method’s sensitivity to network topology/meshedness directly, or an ablation of relational features against a same-information non-relational baseline, which is exactly the comparison this paper’s Sections IV and VII run.

Real-time / latency-aware classification. [4] is the direct comparison point for Section V of this paper – it independently measures the same qualitative gap (fast model, slow pipeline) on a different, larger testbed using an industrial electromagnetic-transient (EMT) simulator, and calls for hardware acceleration where this paper instead finds and fixes a software-level cause for a comparable fraction of the gap.

Zero-day / unseen-attack generalization. The general (non-power-specific) intrusion-detection literature widely reports that supervised detectors trained on known attack families do not reliably generalize to unseen ones; a verified, quantitative comparison figure specific to the power-systems FDIA literature is, at the time of writing, still outstanding work (an earlier draft’s figure here did not survive re-verification against its cited source and was removed rather than repeated).

Field-wide self-assessment of adjacent gaps. A 2025 scoping review of ML applications in power-system protection [19] independently reports, at the level of the field rather than any single paper, two gaps this paper runs into directly: most reviewed studies do not discuss computational latency/real-time constraints at all (Section V), and cross-topology generalization is essentially untested field-wide – reviewed work concentrates on fixed IEEE 14/30/39-bus evaluations with no cross-topology testing (Section VII). This positions those sections as addressing gaps the field has already flagged in itself, not problems specific to this project’s own testbed.

III. TESTBED AND METHODOLOGY

Network. A 5-bus, 0.4kV synthetic microgrid built in pandapower: point of common coupling, a PV bus modeled as a reduced-order grid-following (GFL) inverter, a BESS bus modeled as a reduced-order grid-forming (GFM) inverter, and two load buses, connected by 4 lines. `net.f_hz = 50.0` (verified from the model object, not assumed – this fixes the one-cycle protection-relevant computational target below to 20 ms).

State estimation. Weighted least squares (WLS) AC state estimation via `pandapower.estimation.estimate`, with 23 measurements per cycle: voltage magnitude, and P/Q injection, at all 5 buses, plus P/Q from-side flow on all 4 lines – the same model-driven/data-driven/hybrid-driven taxonomy of state estimation under measurement anomaly

lies surveyed in [20] implicitly underlies this paper’s own `residual_only/prior_only/topology_fusion` split.

Threat model. Two synthetic attack families, both targeting a randomly chosen load bus per scenario:

- *Naive*: a single voltage-magnitude sensor is spoofed by a fixed-range random offset.
- *Model-consistent (“stealth”) FDIA*: following the Liu–Ning–Reiter construction adapted to the AC case, the true state is perturbed at the target bus and the corresponding measurement-space attack vector $a = h(x_{\text{attack}}) - h(x_{\text{true}})$ is computed exactly, so the attack is, by construction, invisible to the chi-square residual test.

A benign hard-negative case (physical_load_disturbance, a 25–70% load step at a randomly chosen bus) is included specifically to be confusable with an attack in raw residual terms.

The prior_only feature set, and exactly how its prior is constructed (the leakage-critical detail). Every replication starts from one randomly sampled base operating point (a load/PV level, shared unmodified across that replication’s 4 scenarios). Before any attack or physical disturbance is applied to build the “true” network for a given scenario, an independent copy of that same base operating point has its load values perturbed by a fresh, independently drawn forecast error ($\mathcal{N}(0, 2.5\%)$ per load, clipped to $\pm 6\%$ in the headline configuration) and is solved through its own, separate AC power flow – this perturbed-and-solved copy is the “prior.” `prior_only` features are the deviation between the WLS state estimate (computed from the scenario’s actual, possibly-attacked measurements) and this prior’s own voltage/angle solution. Critically, the prior is constructed *before*, and entirely independently of, whichever scenario (clean, naive attack, stealth attack, physical disturbance) that replication’s true network becomes – it never sees the true or attacked state, only an independently-perturbed forecast of the same nominal base point. This is what lets stealth attacks, invisible in residual space by construction, become visible at all: the attack is only tuned to be consistent with the WLS residual test, not with an estimate the defender never let it see.

Features. Four feature sets are compared. `residual_only`: conventional WLS measurement residuals. `prior_only`: the prior-deviation features defined above. `residual_plus_prior`: the two combined, with no relational information at all – this is the genuinely topology-free ablation, and the reason it exists is explained next. `topology_fusion`: `residual_plus_prior` plus explicit relational features describing each bus relative to its network neighbors (local-minus-neighbor innovation/residual, incident-edge aggregates).

A feature-labeling bug, and why residual_plus_prior exists. This project’s first pass at these results compared `topology_fusion` only against `residual_only` and `prior_only` individually – never against the union of the two, which is the actual topology-free control a skeptical reviewer would ask for. Building that control surfaced a real bug: `residual_only/prior_only` were

originally defined by matching column-name substrings ("residual", "innovation"), but the explicit relational columns are themselves named after the quantities they compare across neighbors – e.g. `neighbor_mean_residual`, `incident_edge_max_innovation`, `local_minus_neighbor_residual` – so the same substring match silently pulled a large share of every relational column into what were meant to be topology-free baselines (on the 5-bus network: 30 of 65 relational columns into `residual_only`, 35 of 65 into `prior_only`). The practical effect was total: `residual_only` $\cup$ `prior_only`, as originally defined, equaled `topology_fusion` exactly, column for column – so `topology_fusion` could only ever have appeared to beat each baseline *individually* (a comparison of feature *quantity*, more columns vs. fewer), never a comparison that isolated topology-awareness itself, because no topology-free condition using *all* the same non-relational information had ever actually been run. The fix, applied throughout this paper: exclude columns matching explicit relational-name markers (`neighbor_`, `incident_edge_`, `local_minus_`, `local_over_`) from `residual_only/prior_only`, and add their corrected union as the fourth, genuinely topology-free `residual_plus_prior` set. Section IV reports the corrected comparison against the original.

Models. LogisticRegression, RandomForest, and HistGradientBoosting classifiers are compared against 1-feature interpretable-score baselines, with replication-group-safe 70/15/15 train/validation/test splits and group-aware bootstrap confidence intervals.

Sample-size terminology, defined once here to avoid ambiguity below. A *replication* is one independently sampled base operating point (load level, forecast draw); each replication produces exactly 4 *scenarios* – one clean, one naive-attack, one stealth-attack, one physical-disturbance – so N replications yield $4N$ scenarios in total. The 70/15/15 split is applied at the replication level (never at the scenario level, to keep the 4 matched scenarios from one replication from leaking across train/test), so the resulting *test-scenario count* is $0.15N \times 4$, and the *cyber test-scenario count* used for localization (only the 2 attack scenario types have a real target bus) is half of that. Concretely: the headline 5-bus result uses $N=500$ replications $\rightarrow$ 2000 scenarios $\rightarrow \approx 300$ test scenarios; Section VII’s confirmed IEEE 14-bus result uses the same $N=500$ replications (all 500 present) $\rightarrow$ 1993 of the possible 2000 scenarios (seven *physical-disturbance* scenarios, each from a different replication, were excluded because that scenario’s power flow did not converge – the other 3 scenarios in each of those 7 replications are unaffected and included) $\rightarrow$ 300 test scenarios for detection, 150 for localization. Wherever this paper states “ $n=X$ ” without qualification, X is the resulting test-scenario count, not the replication count – the two are only numerically equal here because both studies happened to use 500 replications.

TABLE I
TOPOLOGY_FUSION VS. THE CORRECTED, TOPOLOGY-FREE ABLATION
(BEST MODEL PER CELL; 5-BUS, $n=300$ TEST)

Condition	Detection (bal. acc.)		Localization (top-1)	
	topo. fusion	res.+prior	topo. fusion	res.+prior
Hard	0.950	0.950	0.980	0.993
Harder ($+2\times$ noise)	0.953	0.953	0.980	0.993

IV. RESULTS I: THE TOPOLOGY-FUSION ADVANTAGE DOES NOT SURVIVE ITS OWN ABLATION

An initial near-perfect result was a methodology artifact, not a real effect. The first evaluation of `topology_fusion` reported 1.000 on every detection and localization metric, across three model families, with zero-width bootstrap confidence intervals, on a 72-scenario test set. We treated this as a red flag rather than a result: a defensive coding gap was found and fixed (a feature-exclusion filter for evaluation/label columns that was claimed in a comment but not enforced in code), but re-running showed it had never been the cause – the true cause, already anticipated in the pipeline’s own interpretation logic, was that the original attack scenarios had large, obvious state bias.

Under a realistic attack magnitude, topology_fusion beat prior_only – a real, multi-seed-confirmed, but misleadingly-framed gap. Reducing attack magnitude roughly $3\times$, toward the sensor noise floor, and evaluating at $n=300$ test scenarios gave `topology_fusion` a clear lead over `prior_only` alone: detection 0.950 vs. 0.873 (non-overlapping 95% bootstrap confidence intervals), localization 0.980 vs. 0.780, both stable under a second, independent stress axis (doubled load-forecast uncertainty) and confirmed across four independent seeds (detection gap $+0.089 \pm 0.037$, localization gap $+0.207 \pm 0.032$, sign never flipping). All of this was true, and all of it was reported – but the comparison itself was incomplete: `prior_only` is missing residual information that `topology_fusion` has always had access to, so this gap could not, by construction, distinguish “topology helped” from “having residual information at all helped.”

The fair comparison is a near-exact tie. Table I compares `topology_fusion` against the corrected, genuinely topology-free `residual_plus_prior` ablation (Section III), for the same seed and both stress conditions.

Detection is an exact tie in both conditions; for localization, the topology-free ablation is *ahead* of `topology_fusion`, not behind it. For reference, both feature sets still comfortably beat `prior_only` alone (0.853–0.867 detection, 0.767–0.780 localization) and `residual_only` alone (0.687–0.690 detection, 0.700 localization) – the gain over either single baseline is real and large; it is simply not a topology effect.

The null result replicates across independent random seeds (Fig. 1). Repeating full data generation and evaluation for four independent seeds gives a `topology_fusion`-vs-`residual_plus_prior` detection gap of -0.003 ± 0.010 (range -0.013 to $+0.010$) and a localization gap of $-0.003 \pm$

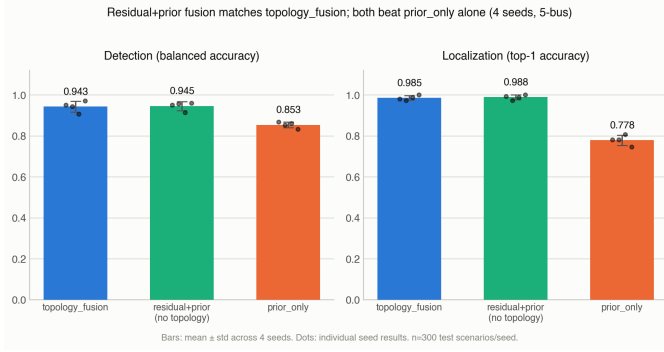


Fig. 1. Topology_fusion vs. the topology-free residual+prior ablation vs. prior_only alone, across 4 independent random seeds (5-bus, hard attack magnitude). Bars: mean $\pm$ std. Dots: individual seed results. $n=300$ test scenarios/seed.

0.007 (range -0.013 to 0.000): both consistent with zero, and – unlike the original, uncorrected topology_fusion-vs-prior_only gap, whose sign never flipped across the same four seeds – the corrected gap’s sign does not hold a consistent direction. This is the signature of noise around zero, not of a real, small effect: the same distinction this project’s own $n=72 \rightarrow n=300$ correction (above) already established, now applied to a different kind of confound.

A structural explanation for the null result. Why would explicit neighbor-comparison features add nothing, given they are computed from genuinely different information – each bus’s residual/innovation relative to its neighbors, not just its own? On this fixed, small (5-bus, 4-line) topology, the answer is close to a linear-algebra fact rather than a coincidence. Predicting each of the 65 relational columns from residual_plus_prior alone, with 5-fold cross-validated ridge regression, gives a mean out-of-fold R^2 of 0.960 (median 1.000, minimum 0.652, 91% of columns above 0.8): a neighbor-mean or incident-edge aggregate over a fixed, small neighbor set is, on a network this size, close to an affine function of the same per-bus values residual_plus_prior already provides in full. A sufficiently flexible classifier (Hist-GradientBoosting, the best model throughout) can reconstruct nearly all of that relational information internally without ever being given it explicitly, which is exactly consistent with Table I’s exact tie. This explanation is specific to a fixed, small, low-redundancy topology, and does not by itself predict the independent IEEE 14-bus null result (Section VII), which uses an unrelated, size-agnostic feature design on a different, larger, meshed network – the two null results are consistent with each other, not mechanistically identical, and a network where relational features are *not* near-linearly redundant given local information remains an open, testable condition for when explicit topology-awareness might actually help.

V. RESULTS II: REAL-TIME FEASIBILITY – DIAGNOSING AND FIXING THE GAP

Following [4]’s finding that fast classifiers can still sit inside a slow end-to-end pipeline, we measured the full per-cycle

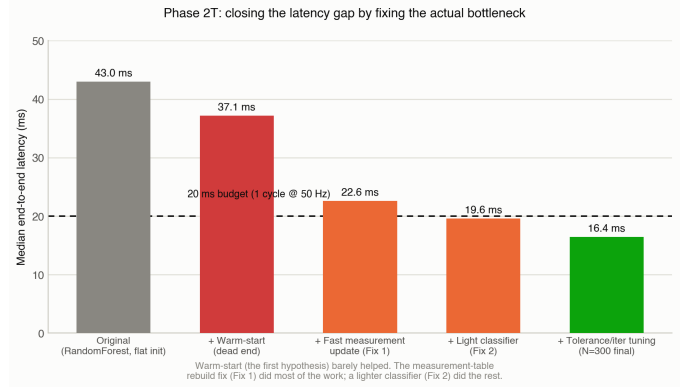


Fig. 2. Latency waterfall from the original pipeline to the final, tuned one. The first hypothesis (warm-start) barely helped; the measurement-table fix did most of the work.

decision pipeline (state estimation $\rightarrow$ feature computation $\rightarrow$ classification) against a one-cycle (20ms at the network’s confirmed 50 Hz) protection-relevant computational target – not a claim that every protective relaying function requires exactly this latency, but the natural reference point for a per-cycle decision loop – with one-time network/model setup excluded, matching how a deployed scheme would actually operate.

Initial measurement: median 42.98 ms, $2.15\times$ over budget, using the original pipeline (RandomForest classifier, tolerance= $1e-7$, maximum_iterations=40, flat state-estimator initialization every cycle).

The first hypothesis (slow convergence from a cold start) was wrong. Warm-starting the state estimator from the previous cycle’s converged solution produced only a $1.06\times$ speedup (37.14ms), ruling out “far from the solution” as the dominant cost.

Profiling found an even split between two causes, only one of them algorithmic. Splitting the state-estimation call into its constituent steps showed the measurement-submission step cost 19.9ms – essentially the same as the estimator’s own solve (19.4ms). An ablation on maximum_iterations (40 $\rightarrow$ 5: no speed change; 40 $\rightarrow$ 3: convergence failure in 19/20 trials) showed the true iteration requirement is roughly 4–5 steps regardless of the 40-iteration cap, ruling out iteration count as the estimator-side cost driver.

The measurement-submission cost was pure, fixable software overhead. The measurement table was being rebuilt from scratch every cycle via a general-purpose element-creation API, called once per channel in a Python loop – even though the table’s structure is identical every cycle; only the values change. Replacing this with a one-time table construction followed by an in-place overwrite of the value column each cycle gave a $1158\times$ speedup on this sub-step (19.4 ms $\rightarrow$ 0.017 ms), verified to produce a bit-identical state-estimation result (max difference 0.0).

Combined with a lighter classifier (LogisticRegression, competitive with RandomForest per Section IV’s accuracy results and $14\times$ faster at inference), the pipeline reaches,

TABLE II
RECALL ON A WITHHELD VS. TRAINED-ON ATTACK TYPE

Attack type	Withheld	In training
Naive single-sensor corruption	0.000	0.893
Model-consistent (stealth) FDIA	0.040	1.000

at $N=300$ trials with 0 convergence failures: **16.41 ms median, 16.95 ms p95, 17.09 ms p99, 21.35 ms max** (Fig. 2) – median, p95, and p99 within the 20 ms budget; a single outlier (out of 300) still exceeds it, traced to the state estimator’s own internal solve-time variability. This benchmark used `topology_fusion`’s 147 features; since Section IV shows `residual_plus_prior`’s 82 features reach the same accuracy, a deployed system could use the smaller feature set for a further (second-order, sub-millisecond) inference saving without any accuracy cost – not separately re-benchmarked here, since the dominant cost by two orders of magnitude is state estimation, not classification, regardless of which of the two feature sets is used.

Positioning. [4] measures the same category of gap (sub-15 ms classification embedded in 50–90 ms pipelines) on an industrial-grade EMT simulator and recommends hardware acceleration. This paper’s finding – that roughly half of a comparable gap traced to an avoidable software inefficiency rather than any algorithmic or numerical cost – suggests pipeline profiling, not only hardware acceleration, deserves consideration first. A second, independent data point supports the “classifier inference itself is cheap” half of this argument: a knowledge-distilled LightGBM detector for virtualized microgrids reports 54–67 ms per 1000 samples (~ 0.06 ms/sample) for inference alone [21] – the same order of magnitude as this paper’s own model-inference sub-step (0.20 ms/sample), and nearly two orders of magnitude below this paper’s state-estimation sub-step (16 ms). Both results point the same way: once a classifier is lightened, the remaining latency budget is spent upstream of the model, not in it.

VI. RESULTS III: THE ADVANTAGE DOES NOT EXTRAPOLATE TO UNSEEN ATTACK TYPES

To test whether Section IV’s detector generalizes beyond the two attack families it was trained on (as opposed to merely discriminating them well), we trained `topology_fusion/HistGradientBoosting` with one entire attack family withheld from training and validation, then measured recall on exactly that withheld family (Table II, Fig. 3). This test uses `topology_fusion` specifically, unaffected by Section IV’s correction (which changes only how `residual_only/prior_only` are defined); whether `residual_plus_prior` shows the same collapse was not independently re-tested, though Section IV’s finding that the two feature sets are behaviorally near-identical in-distribution gives no reason to expect a difference.

The detector recognizes the statistical signature of attack types represented in its training data – including the stealthy

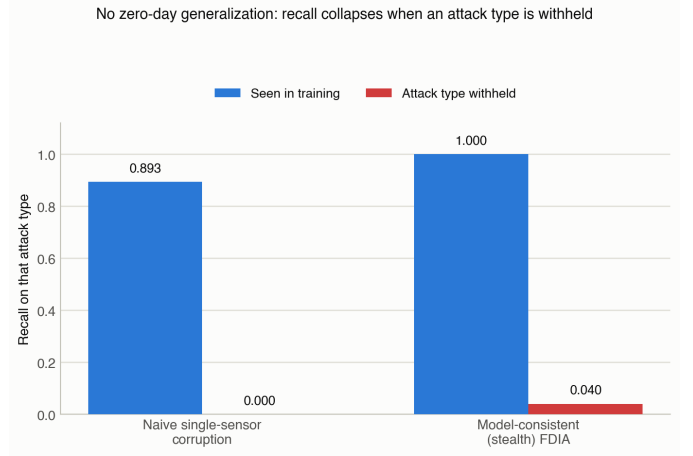


Fig. 3. Recall collapses when an attack type is withheld from training.

one, very well – but does not extrapolate to a mechanism it has not seen. We report this as a hard scope limit on any deployment claim built on Section IV’s results: within this project’s own two-family attack taxonomy, this is a complete absence of zero-day capability, not a partial one. This complements, from the opposite direction, a related generalization gap documented in the fault-classification literature: an XGBoost-based classifier trained and evaluated on the *same* distribution still drops from 96.8% (synthetic data) to 85.2% (physical data) [22] – i.e., even without an unseen attack *type*, a synthetic-to-real distribution shift alone costs over 11 points. Between that gap and this paper’s own 0–4% held-out-type result, generalization – of any kind – looks like the open problem for this class of detector, not an edge case. A deployment-aware zero-day evaluation in the adjacent EV/V2X charging-infrastructure domain [23] is a useful methodological reference for what a more rigorous version of Section VI’s test could look like.

VII. RESULTS IV: AN INDEPENDENT NETWORK SCALE CONFIRMS THE SAME NULL RESULT

Everything in Sections IV–VI uses the custom 5-bus microgrid. We separately tested whether explicit topology-relational features help at a larger, standard scale, escalating sample size across three passes until it matched Section IV’s own 300-scenario test-set size – using a completely independent, size-agnostic feature-engineering design, built before this section’s 5-bus correction and never adjusted to match it.

Test-system selection. The CIGRE European MV benchmark network was tried first and rejected for a concrete engineering reason: it represents internal switches via auxiliary buses (15 pandapower-level buses map to 18 internal power-flow-case buses), which breaks this project’s admittance-matrix-indexing code. IEEE 14-bus has no such mismatch and is, independently, the single most commonly used benchmark in the power-system state-estimation/FDIA literature specifically.

TABLE III
5-BUS VS. IEEE 14-BUS: BEST RESULT PER TASK, AT THE SAME
 $n=300$ -TEST-SCENARIO SIZE ON BOTH SIDES

	Best result
Detection, 5-bus	topology_fusion 0.950 = residual_plus_prior 0.950 (tied; Section IV)
Detection, IEEE-14 (500 repl.)	residual_only/LR 0.750 – best of all, ahead of topology_fusion’s own best (0.737)
Localization, 5-bus	residual_plus_prior 0.993 > topology_fusion 0.980
Localization, IEEE-14 (500 repl.)	topology_fusion 0.560 – narrowly best, but tight (prior_only 0.547, residual_only 0.507); no residual_plus_prior equivalent exists at this scale (different feature design, Section VII)

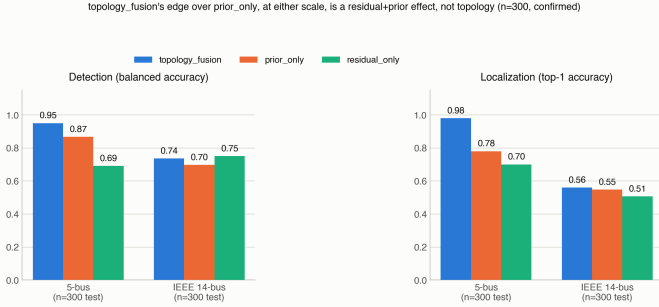


Fig. 4. 5-bus vs. IEEE 14-bus (both at $n=300$ test scenarios): detection and localization by feature set. topology_fusion’s apparent edge over prior_only is present at both scales; Section IV shows it is a residual+prior effect, not a topology one, at 5-bus, and the same pattern (topology_fusion never clearly ahead of a simpler alternative) holds independently at 14-bus.

Method. All WLS/measurement/attack machinery is network-size-agnostic by construction and was reused unmodified. The original fixed per-bus feature columns do not generalize past 5 buses and were replaced with a size-agnostic design: global summary statistics of the residual/innovation arrays, plus, for localization, a trained per-node classifier ranking every bus’s probability of being the attack target. This design predates, and is architecturally unrelated to, the regex-based column split whose bug Section III describes – so it provides an independent check, not a repetition of the same measurement.

Result, three escalating passes, same direction each time (Table III, Fig. 4; replication counts per the terminology defined in Section III). 80 replications (~ 24 test scenarios) was inconclusive by design – the same underpowered regime that had already produced, and been corrected out of, a false reversal in Section IV. 200 replications (60–120 test scenarios) first showed an internally consistent pattern across detection and localization rather than model-to-model noise. 500 replications – yielding a 300-scenario test set, matched exactly to Section IV’s own – did not walk that back; it sharpened it: residual-only detection is the outright leader on IEEE 14-bus, not merely competitive.

Interpretation: two independent measurements, one finding. At first reading, Section IV (5-bus, corrected) and

this section (IEEE 14-bus) look like they address different questions – does topology help, and does that transfer to scale. They do not: both sections independently conclude that explicit topology-relational features carry no measurable advantage over a fair, non-relational baseline, using two unrelated feature-engineering implementations (regex-derived fixed per-bus columns at 5-bus vs. size-agnostic global summaries at 14-bus) built at different points in this project, one of which (14-bus) was never touched by the bug Section III describes. That two independently-built pipelines converge on the same null result is stronger evidence than either alone: it rules out “it was this one feature-engineering implementation” as the explanation, alongside the seed- and stress-condition robustness already shown at 5-bus. This is consistent with, and now sharpens, a 2025 field-wide scoping review’s [19] independent finding that cross-topology generalization is essentially untested across the ML-for-power-protection literature it surveyed – this study is a direct, sample-size-matched, cross-implementation attempt at a gap the field has already named, not a self-inflicted complication and not a preliminary pilot.

VIII. DISCUSSION

Read together, Sections IV–VII converge on one finding, reached twice, independently: *explicit topology-relational features do not measurably improve cyber-physical event detection or localization on this class of problem, once measured against a fair, same-information ablation* – not at a small 5-bus network, and not at a larger, meshed IEEE 14-bus network built with an unrelated feature design. What does help, considerably, is combining residual and prior-innovation information: a real, multi-seed-confirmed ~ 0.09 – 0.21 gain over either alone (Section IV), just not a topology-specific one. Section IV’s redundancy analysis gives this a structural, not just empirical, footing at the smaller scale: on a fixed, small topology, a relational aggregate is close to a linear function of information a flexible classifier already has, so there is little left for an explicit relational feature to contribute. This also sharpens the paper’s remaining open question – not “does topology help,” which this paper answers no to twice, but “on what kind of network would relational aggregates carry information that is *not* already linearly recoverable from local features,” which Section IV’s method (an out-of-fold redundancy check, cheap to run on any candidate network before training a single classifier) can test directly, before running the full detection/localization pipeline at all.

The paper’s original framing – “does topology help, and where does that stop” – was itself built on an assumption (that residual_only/prior_only were topology-free) that turned out to be false by construction, not disproven by measurement; the more important finding here is not a scale boundary on a real effect, but the absence of the effect originally claimed, caught only by building the specific ablation a skeptical reviewer would have asked for regardless. We think this pattern – check the suspiciously good result before trusting it, then build the control condition the result was never actually

tested against – is itself a methodological point worth stating explicitly, and a cheap one: the ablation that overturned this paper’s own headline claim required no new data collection, only a corrected column filter and one additional feature set trained on data already in hand.

IX. LIMITATIONS

- 1) **Scale:** Section VII’s IEEE 14-bus study independently cross-validates Section IV’s corrected 5-bus finding using an unrelated feature-engineering design, but only two network topologies have been tested overall, and neither uses a real (non-synthetic) network or load profile.
- 2) **No zero-day generalization** (Section VI): detection/localization numbers hold only for attack types represented in training, and this was verified for `topology_fusion` specifically, not independently re-run for `residual_plus_prior`.
- 3) **Single testbed family, two attack types:** restricting which measurements a stealth attack compromises is a physical consequence of the AC-consistent attack construction, not a tunable knob, and was not varied.
- 4) **Latency results** (Section V) characterize one Python/pandapower implementation on one machine, not a hardware or RTOS claim, and were not re-benchmarked on the smaller `residual_plus_prior` feature set (Section V argues this would not change the conclusion, since state estimation, not classification, dominates the budget, but this is an argument, not a re-measurement).
- 5) **Related-work verification is partial:** of ~56 sources surveyed during this project, 40 were read past search-summary level; two invented or misattributed figures were caught and corrected during that process, which is grounds for treating any remaining unstarred source as provisional.

X. CONCLUSION

On both a small inverter-dominated microgrid and a larger, meshed IEEE 14-bus network, explicit topology-relational features show no measurable advantage over a fair, same-information ablation (residual and prior-innovation features combined) for cyber-physical event detection or localization – a finding reached independently at both scales, with two unrelated feature-engineering implementations, and confirmed across two attack-stress conditions and four independent seeds at the smaller scale. This null result was not the project’s starting hypothesis: an initial, multi-seed-validated positive result (`topology_fusion` beating `prior_only` alone) was real, but built on an incomplete comparison, exposed only once a genuinely topology-free control condition was constructed and a feature-labeling bug it revealed was corrected. Separately, the resulting decision pipeline can be brought within a one-cycle protection-relevant computational target once its actual bottleneck – a software inefficiency, not the model or the estimator’s numerics – is found and fixed, and the detector

does not extend to attack types unseen in training. We think the corrected, largely-negative version of this result is more useful, and more publishable, than the original positive one would have been: it identifies a specific methodological trap (comparing an information-rich feature set against baselines that were not actually information-poor) that is easy to fall into and cheap to check for, in exactly the topology-aware / graph-based detection literature this paper set out to contribute to.

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Experiment design, data generation, and evaluation for this work were carried out with AI-assisted software engineering and literature review (Claude, Anthropic); all reported quantitative claims trace to scripts and result files in the accompanying repository, and all literature claims marked as independently read in `RELATED_WORK.md` were verified against primary sources rather than taken from search summaries alone. The feature-labeling bug central to this paper’s corrected finding (Section III) was caught during an AI-assisted code review while implementing the `residual_plus_prior` ablation requested as a next step in this project’s own research process; the correction and re-analysis in Sections IV and VII were completed, and the original positive framing withdrawn, before this manuscript was finalized.

REFERENCES

- [1] N. Liaquat, N. Chihaoui, M. Nasir, A. S. S. Mufleh, S. M. Alshraah, and A. Waleed, “Protection challenges and emerging solutions in renewable-integrated microgrids: a critical review,” *Energy Informatics*, vol. 9, no. 1, p. 24, 2026.
- [2] Y. Liu, P. Ning, and M. K. Reiter, “False data injection attacks against state estimation in electric power grids,” *ACM Transactions on Information and System Security*, vol. 14, no. 1, pp. 13:1–13:33, 2011.
- [3] CERT Polska, “Follow-up analysis of the 29 december 2025 energy sector incident,” CERT Polska, NASK, Tech. Rep., 8 2026, <https://cert.pl/en/posts/2026/08/incident-follow-up-report-energy-sector-2025/>.
- [4] E. Abukhousa, S. Zonouz, and A. P. S. Meliopoulos, “Latency-aware deep learning benchmark for real-time cyber-physical attack and fault classification in inverter-dominated power grids,” *arXiv preprint arXiv:2605.17256*, 2026.
- [5] A. Anwar, A. N. Mahmood, and M. Pickering, “Modeling and performance evaluation of stealthy false data injection attacks on smart grid in the presence of corrupted measurements,” *arXiv preprint arXiv:1605.06180*, 2016.
- [6] M. Du, G. Pierrou, and X. Wang, “Targeted false data injection attack against dc state estimation without line parameters,” *arXiv preprint arXiv:2102.11896*, 2021.
- [7] S. Zhao, W. Luo, Q. Shu, and F. Xu, “Controllable blind ac fdia via physics-informed extrapolative avae,” *Sensors*, vol. 25, no. 3, 2025.
- [8] S. Falas, M. Asprou, C. Konstantinou, and M. K. Michael, “Learning without adversarial training: A physics-informed neural network for secure power system state estimation under false data injection attacks,” *arXiv preprint arXiv:2604.22784*, 2026.
- [9] X. Li, C. Xiao, J. Cohen, A. Elyashar, Y. Weng, and R. Puzis, “Physically consistent null space alignment for detection of low-magnitude false data injection attacks,” *arXiv preprint arXiv:2606.08473*, 2026.
- [10] X. Niu, J. Li, and J. Sun, “Dynamic detection of false data injection attack in smart grid using deep learning,” *arXiv preprint arXiv:1808.01094*, 2018.
- [11] J. Li, Y. Yang, J. S. Sun, K. Tomsovic, and H. Qi, “Towards adversarial-resilient deep neural networks for false data injection attack detection in power grids,” *arXiv preprint arXiv:2102.09057*, 2021.
- [12] B. Xu, F. Guo, C. Wen, R. Deng, and W.-A. Zhang, “Detecting false data injection attacks in smart grids with modeling errors: A deep transfer learning based approach,” *arXiv preprint arXiv:2104.06307*, 2021.

- [13] S. Moshtagh, B. Azimian, M. Golgol, and A. Pal, "Topology-aware graph neural network-based state estimation for pmu-unobservable power systems," *arXiv preprint arXiv:2506.03493*, 2025.
- [14] D. Suri and M. Mangal, "Powergnn: A topology-aware graph neural network for electricity grids," *arXiv preprint arXiv:2503.22721*, 2025.
- [15] A. Yaniv, K. Golinski, and C. Goebel, "Robust and interpretable graph neural networks for power systems state estimation," *arXiv preprint arXiv:2603.23357*, 2026.
- [16] H. Jahangir, S. Lakshminarayana, and C. Maple, "Localizing load-altering attacks against power grids using deep capsule nets," *arXiv preprint arXiv:2209.02809*, 2022.
- [17] Y. Chen, S. Lakshminarayana, and F. Teng, "Localization of coordinated cyber-physical attacks in power grids using moving target defense and deep learning," *arXiv preprint arXiv:2207.12339*, 2022.
- [18] D. Li, N. Gebräel, K. Paynabar, and A. P. S. Meliopoulos, "An online approach to cyberattack detection and localization in smart grid," *arXiv preprint arXiv:2102.11401*, 2021.
- [19] J. Oelhaf, G. Kordowich, M. Pashaei, C. Bergler, A. Maier, J. Jäger, and S. Bayer, "A scoping review of machine learning applications in power system protection and disturbance management," *arXiv preprint arXiv:2509.09053*, 2025.
- [20] T. Lin, J. Zhang, Z. Lin, J. Li, C. Li, and X. Xu, "State estimation of power systems under measurement anomalies," *Energies*, vol. 19, no. 3, p. 632, 2026.
- [21] O. C. Ogiesoba-Eguakun and S. Rath, "Cyberattack detection in virtualized microgrids using lightgbm and knowledge-distilled classifiers," *arXiv preprint arXiv:2601.03495*, 2026.
- [22] M. Senyuk, S. Beryozkina, I. Zicmane, M. Safaraliev, V. Klassen, and F. Kamalov, "Bulk low-inertia power systems adaptive fault type classification method based on machine learning and phasor measurement units data," *Mathematics*, vol. 13, no. 2, p. 316, 2025.
- [23] H. A. Sakr, A. A. El-Douh, M. Lapina, V. Lapin, B. Senapati, and M. I. El-Afifi, "Explainable and deployment-aware zero-day intrusion detection for cloud-level backend and management ecosystems in ev/v2x cyber-physical systems," *Computers*, vol. 15, no. 9, p. 599, 2026.